

One world context, many views



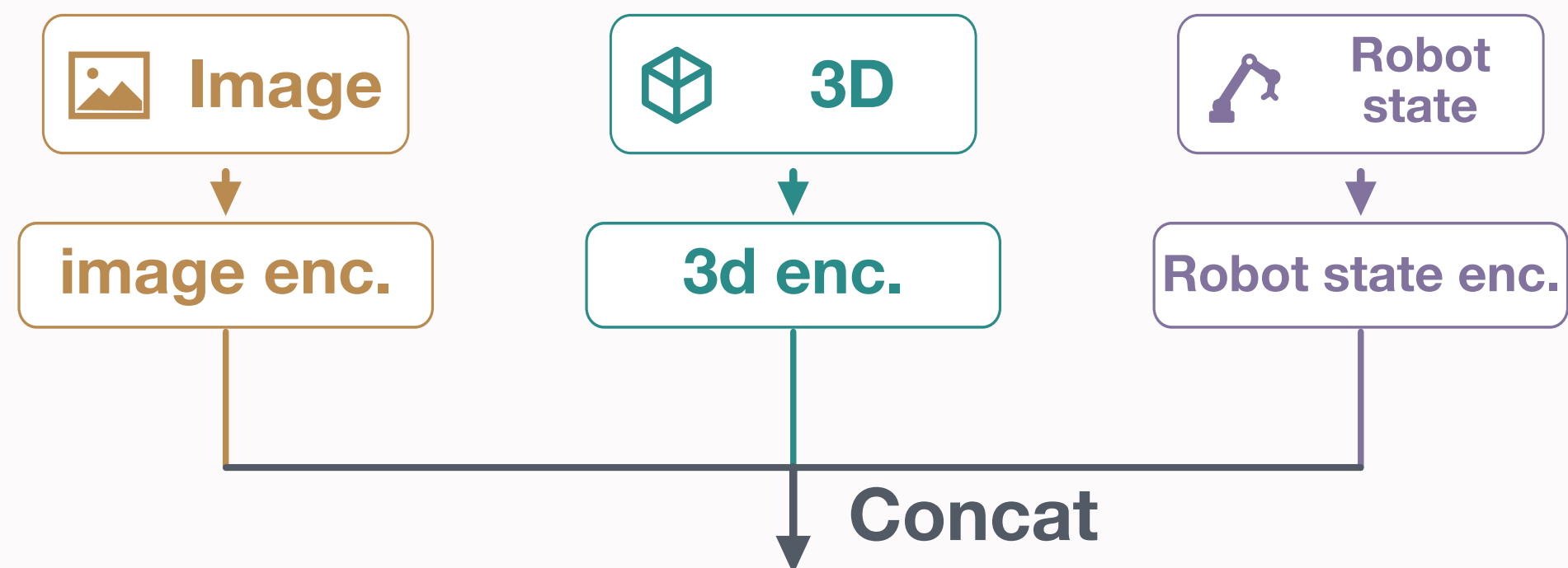
Task input x

VLM Text + Image
What is beside the blue book?

VLA Text + Image + Robot state
Pick up the red mug.

Existing multimodal CoT

Encode, then concatenate



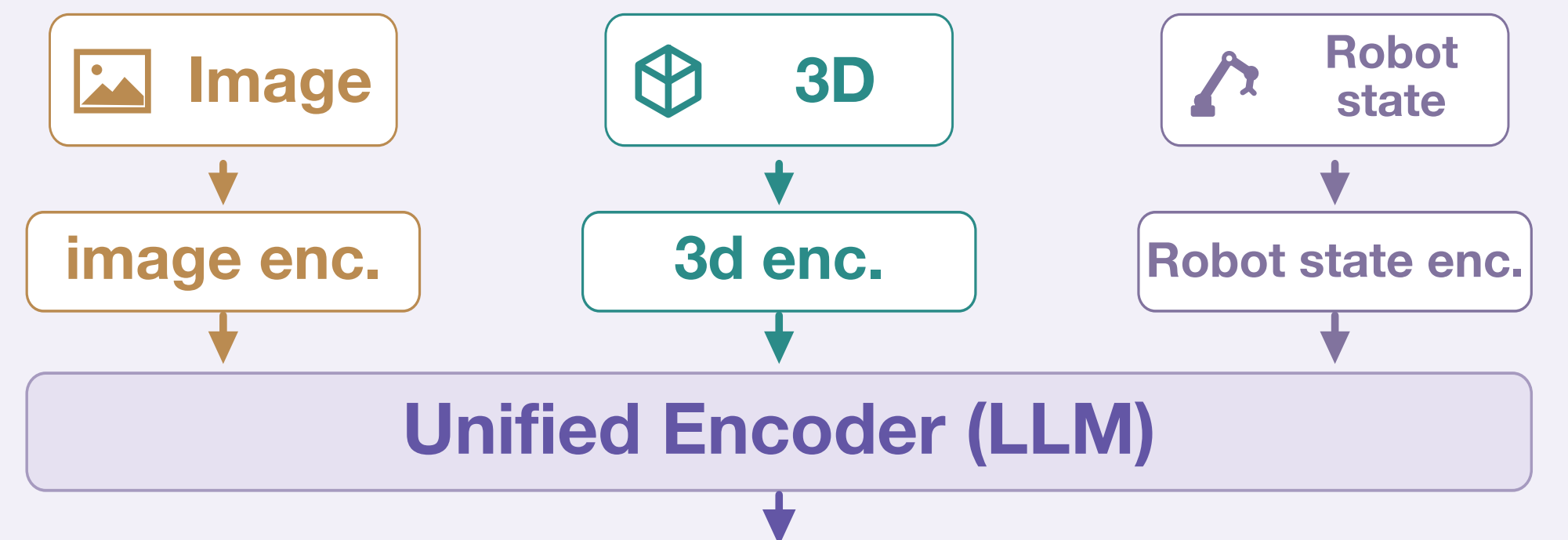
Reasoning chain



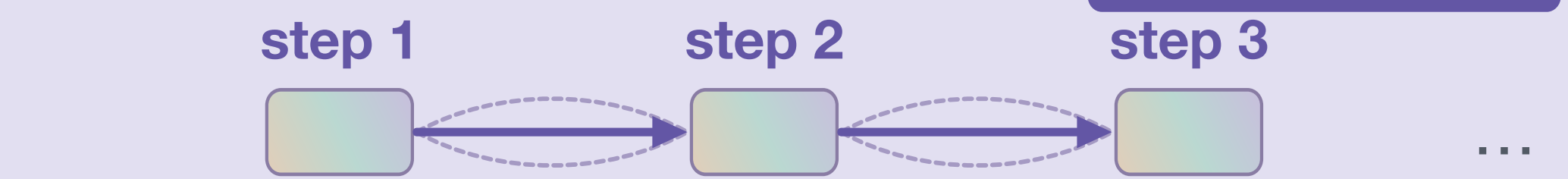
modality-switching gap

Uni-LaDiR (Ours)

Encode into one reasoning space



Reasoning chain



Unified latent space

Output y

Task-dependent

VLM · Text answer

A red mug.

VLA · Action sequence

$[a_1, a_2, \dots]$

